

DIGITAL SAT — MATH

Theory Companion — Part 1

Algebra & Data Analysis — Theory & Worked Problems

- 13 skills, one representative Hard/Very Hard worked problem per skill
- Algebra: Lines • Systems • Inequalities • Absolute Value • Exponents • Radicals
- Problem-Solving & Data Analysis: Ratios & Proportions • Probability • Unit Conversions • Statistics • Data Displays • Scatterplots • Sampling
- Companion to the full 13-skill, 104-question Math Theory Study Companion

The 13 Skills

1. Lines
2. Systems of Linear Equations
3. Linear Inequalities
4. Absolute Value
5. Exponent Rules
6. Radicals, Rational Exponents & Scientific Notation
7. Ratios & Proportions
8. Probability
9. Unit Conversions & Rates
10. Statistics — Center & Spread
11. Data Displays — Histograms, Dot Plots & Box Plots
12. Scatterplots
13. Sampling, Bias & Margin of Error



PART 1

Algebra

Lines • Systems of Linear Equations • Linear Inequalities • Absolute Value • Exponent Rules • Radicals, Rational Exponents & Scientific Notation

Core Rules

- Slope Formula — $m = (y_2 - y_1) / (x_2 - x_1)$
- Slope-Intercept Form — $y = mx + b$
- Point-Slope Form — $y - y_1 = m(x - x_1)$
- Standard Form — $Ax + By = C \rightarrow \text{slope} = -A/B, \text{ y-intercept} = C/B$
- Parallel & Perpendicular Lines — Parallel: $m_1 = m_2$. Perpendicular: $m_1 \times m_2 = -1$

Watch Out: Never count grid squares to estimate slope from a graph — the axes are often scaled differently. Always use the actual coordinate values.

Segment AB has endpoints A(-2, 5) and B(6, 1). Line m is the perpendicular bisector of segment AB. What is the y-intercept of line m?

Formula / Rule Used: Midpoint formula and negative reciprocal slopes (perpendicular bisector).

Work through it, then advance for the full step-by-step solution.

Step 1: Find the midpoint of segment AB: $M = ((-2+6)/2, (5+1)/2) = (4/2, 6/2) = (2, 3)$.

Step 2: Find the slope of AB: $m_{AB} = (1 - 5)/(6 - (-2)) = -4/8 = -1/2$.

Step 3: Line m is the perpendicular bisector, so its slope is the negative reciprocal of $-1/2$, which is 2.

Step 4: Line m passes through the midpoint (2, 3) with slope 2. Use point-slope form: $y - 3 = 2(x - 2)$.

Step 5: Simplify: $y - 3 = 2x - 4$, so $y = 2x - 1$.

Step 6: The y-intercept occurs where $x = 0$: $y = 2(0) - 1 = -1$.

Answer: y-intercept = -1, so line m crosses the y-axis at (0, -1).

Tip: For a perpendicular bisector problem, always compute the midpoint AND the perpendicular slope before writing the equation -- both pieces are required.

Trap: A common error is building the bisector line through one of the original endpoints (A or B) instead of the midpoint, which produces a different, incorrect y-intercept.

Tempting Wrong Answers — and Exactly What Went Wrong

y-intercept = 9

Used endpoint A(-2, 5) instead of the midpoint (2, 3) as the point for the bisector line, keeping the correct perpendicular slope of 2 but building the equation through the wrong point.

y-intercept = 4

Used the slope of segment AB itself (-1/2) instead of its negative reciprocal (2) when writing the bisector's equation through the midpoint.

y-intercept = 7

Made a sign error finding the slope of AB, computing $(1-5)/(6+2)$ as $4/8 = 1/2$ instead of $-4/8 = -1/2$, which flipped the perpendicular slope to -2 instead of 2.

Shortcut: Since only the y-intercept is needed, you can skip fully simplifying to standard form -- just solve the point-slope equation for y and read off the constant term directly.

Desmos Tip: Plot A(-2,5) and B(6,1), graph the segment, then graph $y = 2x - 1$ and confirm it crosses AB at a right angle exactly at the midpoint (2,3).

Core Rules

- Substitution — Isolate one variable in one equation, substitute it into the other, solve, then back-substitute. Best when a variable already has coefficient 1.
- Elimination — Multiply one or both equations so a variable's coefficients become opposites, then add the equations to cancel that variable.
- Setting Equal — $f(x) = g(x)$
- Solving Directly for an Expression (shortcut) — If asked for a combined value like $2x + 3y$ rather than x and y individually, check whether adding/subtracting/scaling the two equations produces that expression directly — this is almost always faster than solving the full system.
- Number of Solutions — One: $m_1 \neq m_2$. None: $m_1 = m_2$, $b_1 \neq b_2$. Infinite: $m_1 = m_2$, $b_1 = b_2$.

Watch Out: For a 'no solution' or 'infinite solutions' question with an unknown constant, use the coefficient-ratio shortcut instead of solving the whole system — it turns the problem into a simple proportion.

Consider the system $2x = 11 - 5y$ and $ky = 30 - 6x$, where k is a constant. For what value of k does this system have no solution?

Formula / Rule Used: No-solution condition for a linear system: $A1/A2 = B1/B2$ does not equal $C1/C2$ in standard form $Ax + By = C$.

Work through it, then advance for the full step-by-step solution.

Step 1: Rewrite the first equation in standard form: $2x + 5y = 11$.

Step 2: Rewrite the second equation in standard form: $6x + ky = 30$.

Step 3: For no solution, the coefficient ratios must match: $2/6 = 5/k$.

Step 4: Cross-multiply: $2k = 30$, so $k = 15$.

Step 5: Check that the constant ratio is genuinely different (required for no solution, not infinite solutions): $C1/C2 = 11/30$, while $A1/A2 = 2/6 = 1/3 = 10/30$.

Step 6: Since $11/30$ does not equal $10/30$, the lines are parallel but distinct at $k = 15$, confirming no solution.

Answer: $k = 15$

Tip: Before comparing ratios, rearrange every equation into $Ax + By = C$ form -- don't try to compare coefficients while variables and constants are mixed on both sides.

Trap: A student who finds $k = 15$ from the coefficient ratio alone often stops and assumes matching slopes automatically means infinitely many solutions; they must also check that the constant ratio ($11/30$) differs from the coefficient ratio ($1/3$) to confirm this is actually a no-solution case.

Tempting Wrong Answers — and Exactly What Went Wrong

$k = -15$

Rearranged $ky = 30 - 6x$ incorrectly as $6x - ky = 30$ (sign error on the y-term) instead of $6x + ky = 30$, which flips the sign in the ratio and produces $k = -15$ instead of $k = 15$.

$k = 12/5$

Mispaired the coefficients when writing the ratio, using $5/6 = 2/k$ (y-coefficient of equation 1 over x-coefficient of equation 2) instead of the correct $2/6 = 5/k$, mixing up which coefficients correspond to which equation.

$k = 5/3$

Cross-multiplied the proportion $2/6 = 5/k$ backwards, computing $2 \times 5 = 6 \times k$ instead of $2 \times k = 6 \times 5$, giving $k = 5/3$ instead of 15.

Shortcut: Once both equations are in standard form, skip solving the system entirely -- just cross-multiply the x- and y-coefficient ratio ($2/6 = 5/k$) to get $k = 15$ in one line.

Desmos Tip: Graph $2x + 5y = 11$ and $6x + 15y = 30$ (solved for y) in Desmos and confirm the two lines are visibly parallel and never touch.

Core Rules

- The Flip Rule — Multiply/divide by a negative $\rightarrow$ reverse the sign.
- Compound (Three-Part) Inequalities — $a < x < b$
- Graphing in the xy-Plane — Solid line for $\leq / \geq$ (boundary included); dashed line for $< / >$ (boundary excluded). Shade above for $y > / \geq$, below for $y < / \leq$, or test a point like (0,0).
- Systems of Inequalities — Graph each inequality; the solution is the overlapping shaded region. To check a point, substitute it into every inequality.
- Word Problems — 'at least/no less than' $\rightarrow \geq$. 'at most/no more than' $\rightarrow \leq$.

Watch Out: Real-world quantities (people, tickets, boxes) often force you to round the algebraic answer to the nearest valid integer — always re-read what's being asked (max vs. min) before rounding.

Solve the compound inequality for x , and express the solution set using inequality notation: $-3 < (2 - 5x)/3 \leq 4$.

Formula / Rule Used: Compound (three-part) inequality: perform the same operation on all three parts, reversing both inequality signs and swapping the endpoint order when multiplying or dividing by a negative number.

Work through it, then advance for the full step-by-step solution.

Step 1: Multiply all three parts by 3 to clear the denominator: $-3 \times 3 < 2 - 5x \leq 4 \times 3$, giving $-9 < 2 - 5x \leq 12$.

Step 2: Subtract 2 from all three parts: $-9 - 2 < -5x \leq 12 - 2$, giving $-11 < -5x \leq 10$.

Step 3: Divide all three parts by -5. Since -5 is negative, reverse both inequality signs and flip the order of the parts: $-11/-5 > x \geq 10/-5$.

Step 4: Simplify each side: $-11/-5 = 11/5$ and $10/-5 = -2$, so the inequality reads $11/5 > x \geq -2$.

Step 5: Rewrite in standard increasing order (smallest to largest, left to right): $-2 \leq x < 11/5$.

Step 6: Check an endpoint: at $x = -2$, $(2 - 5(-2))/3 = (2 + 10)/3 = 12/3 = 4$, which satisfies ≤ 4 with equality, confirming $x = -2$ is correctly included.

Step 7: Check the other endpoint: at $x = 11/5$, $(2 - 5(11/5))/3 = (2 - 11)/3 = -9/3 = -3$, which is NOT strictly greater than -3 (it equals -3), confirming $x = 11/5$ is correctly excluded since the original inequality was strict ($>$) on that side.

Answer: $-2 \leq x < 11/5$

Tip: Clear the fraction first by multiplying every part by the denominator, then isolate x exactly as with a normal compound inequality, remembering to flip both inequality signs and reverse the order of the endpoints whenever you multiply or divide by a negative number.

Trap: A student who divides by -5 but forgets to also swap which endpoint is 'less than' and which is 'greater than' will report the answer as $11/5 \leq x < -2$, which is an empty (impossible) set since $11/5$ is greater than -2.

Tempting Wrong Answers — and Exactly What Went Wrong

$$11/5 \leq x < -2$$

Flipped the inequality symbols correctly when dividing by -5 but forgot to also swap the left-right order of the two endpoints, producing an impossible set where the 'lower' bound is larger than the 'upper' bound.

$$-2/5 \leq x < 1$$

Skipped clearing the fraction (never multiplied all three parts by 3) and instead subtracted 2 and divided by -5 directly on the un-cleared expression $(2 - 5x)/3$, shifting every boundary value.

$$-2 < x \leq 11/5$$

Correctly flipped the inequality direction and endpoint order but attached the strict '<' symbol to the wrong endpoint, swapping which side is open and which is closed.

Shortcut: No shortcut beats clearing the fraction first -- trying to divide by 3 termwise before combining like terms just recreates the same fraction later, so multiplying through immediately is already the fastest path.

Desmos Tip: Graph $y = (2 - 5x)/3$ along with the horizontal lines $y = -3$ and $y = 4$, then read off the x-interval where the slanted line stays between the two horizontal lines (open circle at $x = 11/5$, closed circle at $x = -2$).

Core Rules

- Solving $|X| = a$ — If $a \geq 0$: $X = a$ or $X = -a$. If $a < 0$: no solution.
- Absolute Value Inequalities — $|X| < a \rightarrow -a < X < a$. $|X| > a \rightarrow X < -a$ or $X > a$ ($a > 0$)
- $|A| = |B|$ — $A = B$ or $A = -B$
- Extraneous Solutions — Whenever a variable appears both inside and outside the absolute value bars (or gets squared), you must check every solution in the ORIGINAL equation — one branch often fails.
- Distance Interpretation — $|x - a|$ = distance between x and a

Watch Out: If isolating the absolute value ever produces $|X| = (\text{negative number})$, stop — there is no solution. Don't split into cases anyway.

The inequality $5|x - 3| - 7 > k$ has solution set $x < -1$ or $x > 7$. What is the value of k ?

Formula / Rule Used: $|X| > a$ ($a > 0$) splits into $X > a$ or $X < -a$.

Work through it, then advance for the full step-by-step solution.

Step 1: Isolate the absolute value expression first: add 7 to both sides to get $5|x-3| > k + 7$.

Step 2: Divide both sides by 5: $|x - 3| > (k+7)/5$.

Step 3: An inequality of the form $|X| > a$ splits into $X > a$ or $X < -a$, so $x - 3 > (k+7)/5$ or $x - 3 < -(k+7)/5$.

Step 4: Rewrite each branch by adding 3: $x > 3 + (k+7)/5$, or $x < 3 - (k+7)/5$.

Step 5: Match the given upper boundary $x > 7$ to the first branch: $3 + (k+7)/5 = 7$, so $(k+7)/5 = 4$, giving $k + 7 = 20$, so $k = 13$.

Step 6: Check against the given lower boundary $x < -1$: $3 - (k+7)/5 = -1$, so $(k+7)/5 = 4$ again, giving the same $k = 13$.

Step 7: Both boundaries agree, so $k = 13$.

Answer: $k = 13$

Tip: When you're given the finished solution set of an absolute value inequality and asked for a hidden constant, rebuild the general $|x - \text{center}| > \text{distance}$ form and match its two boundary numbers to the given ones algebraically.

Trap: A rushed student plugs $x = 7$ or $x = -1$ directly into the original inequality as if it were an equation, but forgets to first isolate $|x-3|$ by undoing the -7 and dividing by 5, and often reports $k = 7$ by ignoring the coefficient of 5 entirely.

Tempting Wrong Answers — and Exactly What Went Wrong

k = 20

Dropped the -7 when isolating, treating $5|x-3|-7 > k$ as $5|x-3| > k$, which gives $|x-3| > k/5$; matching the boundary $x=7$ then gives $k/5 = 4$, so $k = 20$ instead of 13.

k = -3

Correctly isolated the absolute value but forgot to divide by 5, using $3 + (k+7) = 7$ instead of $3 + (k+7)/5 = 7$, giving $k = -3$.

k = -31/5

When solving $(k+7)/5 = 4$, cross-multiplied backwards, writing $k+7 = 4/5$ instead of $k+7 = 4(5) = 20$, producing a wrong fractional value for k.

Shortcut: Since 3 is exactly the midpoint of the two boundary values -1 and 7, the 'radius' distance is $(7 - (-1))/2 = 4$, so $(k+7)/5 = 4$ can be written immediately without deriving both boundary equations separately.

Desmos Tip: Graph $y = 5 \cdot \text{abs}(x-3) - 7$ and the horizontal line $y = k$, then adjust k until the shaded region's edges land exactly at $x = -1$ and $x = 7$ to visually confirm $k = 13$.

Core Rules

- Product & Quotient Rules — $a^m \cdot a^n = a^{m+n}$ $a^m / a^n = a^{m-n}$
- Power Rules — $(a^m)^n = a^{mn}$ $(ab)^n = a^n \cdot b^n$ $(a/b)^n = a^n / b^n$
- Zero & Negative Exponents — $a^0 = 1 (a \neq 0)$ $a^{-n} = 1/a^n$
- Fractional Exponents — $a^{m/n}$ = n-th root of a^m = n-th root of (a^m)
- Solving Exponential Equations — If $a^m = a^n$ (same base), then $m = n$.

Watch Out: $(a + b)^n$ is NEVER $a^n + b^n$ — you cannot distribute an exponent over addition or subtraction.

Simplify to a single term with positive exponents only: $[(2x^{-3}y^2)^3 \cdot (4x^5y^{-1})^{-1}] / (x^2y^{-3})^0$

Formula / Rule Used: Power of a product, product rule, negative exponents, and the zero-exponent rule together.

Work through it, then advance for the full step-by-step solution.

Step 1: Recognize that any nonzero expression raised to the power 0 equals 1, so the denominator $(x^2y^3)^0 = 1$ -- it does not need to be expanded.

Step 2: Apply the power rule to the first factor: $(2x^3y^2)^3 = 2^3 \cdot x^{3 \cdot 3} \cdot y^{2 \cdot 3} = 8x^9y^6$.

Step 3: Apply the power rule to the second factor: $(4x^5y^{-1})^{-1} = 4^{-1} \cdot x^{5 \cdot -1} \cdot y^{-1 \cdot -1} = (\frac{1}{4})x^{-5}y$.

Step 4: Multiply the two results using the product rule (add exponents of like bases): $8x^9y^6 \cdot (\frac{1}{4})x^{-5}y = ((8 \cdot \frac{1}{4}) \cdot x^{9-5} \cdot y^{6+1}) = 2x^4y^7$.

Step 5: Divide by the denominator, which is 1, so the expression is unchanged: $(2x^4y^7)/1 = 2x^4y^7$.

Step 6: Rewrite with positive exponents only: $2x^4y^7 = (2y^7)/x^{-4}$.

Answer: $(2y^7)/x^{14}$

Tip: When a factor in the expression is raised to the 0 power, simplify it to 1 immediately instead of expanding it -- it never affects the rest of the problem.

Trap: A student who doesn't recognize $(x^2y^3)^0 = 1$ wastes time expanding it, or worse, mistakenly distributes the 0 as if it were a real exponent to multiply through, instead of just replacing the whole denominator with 1.

Tempting Wrong Answers — and Exactly What Went Wrong

$$2x^{-14}y^7$$

Correctly combined all exponents but forgot the last step of rewriting negative exponents as positive -- stopped before flipping x^{-14} to $1/x^{14}$.

$$(2y^5)/x^{14}$$

Mis-applied the power rule to $(4x^3y^1)^{-1}$, treating $y^{1 \cdot -1}$ as y^1 instead of y^{-1} , which lowers the final y exponent from 7 to 5.

$$-(32y^7)/x^{14}$$

Misinterpreted the negative exponent on the coefficient, computing 4^{-1} as -4 instead of the reciprocal $\frac{1}{4}$, which corrupts the numeric coefficient in the final answer.

Shortcut: Besides recognizing the zero-power factor as 1, there's no shortcut here -- the fastest path is still applying the power rule to each bracket separately before combining, exactly as shown.

Desmos Tip: Pick simple values like $x=2$, $y=1$, evaluate the original expression and the simplified answer $(2y^7)/x^{14}$ in Desmos's calculator, and confirm they give the same number.

Core Rules

- Radical-Exponent Conversion — n -th root of $(a^m) = a^{m/n}$
- Simplifying Radicals — $\sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$
- Combining Radicals with Exponents — Convert every term to a fractional exponent with a common denominator, then apply the product/quotient rule, then convert back if needed.
- Scientific Notation — $N \cdot 10^n$, where $1 \leq N < 10$

Watch Out: After any scientific-notation multiplication or division, check that your coefficient is still between 1 and 10 — if not, shift the decimal and adjust the exponent by 1.



Write $\sqrt[3]{(x^2) \cdot \sqrt{(x^3)}} / x^{\frac{1}{2}}$ as a single power of x (assume $x > 0$), and simplify to its lowest exponent.

Formula / Rule Used: Convert radicals to rational (fractional) exponents with a common denominator, then apply the product and quotient rules for exponents.

Work through it, then advance for the full step-by-step solution.

Step 1: Convert $\text{cbrt}(x^2)$ to a fractional exponent: $x^{\frac{2}{3}}$.

Step 2: Convert $\sqrt{(x^3)}$ to a fractional exponent: $x^{\frac{3}{2}}$.

Step 3: Multiply these using the product rule for exponents (add exponents): $x^{\frac{2}{3}} \cdot x^{\frac{3}{2}} = x^{\frac{2}{3} + \frac{3}{2}}$.

Step 4: Find a common denominator of 6 to add the exponents: $\frac{2}{3} = \frac{4}{6}$ and $\frac{3}{2} = \frac{9}{6}$, so $\frac{2}{3} + \frac{3}{2} = \frac{4}{6} + \frac{9}{6} = \frac{13}{6}$.

Step 5: The expression so far is $x^{\frac{13}{6}}$.

Step 6: Divide using the quotient rule for exponents (subtract exponents): $x^{\frac{13}{6}} / x^{\frac{1}{6}} = x^{\frac{13}{6} - \frac{1}{6}} = x^{\frac{12}{6}}$.

Step 7: Simplify the exponent: $\frac{12}{6} = 2$.

Step 8: Final simplified form: x^2 .

Answer: x^2

Tip: Convert every root to a fractional exponent immediately — mixing cube roots and square roots directly is what causes errors, but once everything is written as $x^{p/q}$, the problem is just exponent addition and subtraction.

Trap: A student might try to combine $\sqrt{(x^3)}$ and $\text{cbrt}(x^2)$ by rewriting both as 6th roots and multiplying radicands directly, which can work but is far more error-prone; the most common slip is treating the division by $x^{\frac{1}{6}}$ as another multiplication instead of a subtraction, giving $x^{\frac{14}{6}}$ instead of the correct $x^{\frac{12}{6}}$.

Tempting Wrong Answers — and Exactly What Went Wrong

$x^{\frac{1}{6}}$

Treated the division by $x^{\frac{1}{6}}$ as multiplication instead of subtraction, computing $\frac{13}{6} + \frac{1}{6} = \frac{14}{6} = \frac{7}{3}$ instead of $\frac{13}{6} - \frac{1}{6} = \frac{12}{6}$.

$x^{\frac{1}{6}}$

Correctly converted and multiplied the two radicals to get $x^{\frac{1}{6}}$, but forgot to divide by $x^{\frac{1}{6}}$ at all, stopping one step early.

$x^{\frac{1}{6}}$

Added the exponents $\frac{2}{3}$ and $\frac{3}{2}$ by adding numerators and denominators straight across ($\frac{2}{3} + \frac{3}{2}$ treated as $\frac{5}{5} = 1$) instead of finding a common denominator of 6, corrupting the exponent before the final subtraction.

Shortcut: Since all three exponents already share a natural common denominator of 6, you can rewrite the whole expression as a single 6th root from the start: sixth root of $((x^4 \cdot x^9)/x^1) =$ sixth root of $x^{12} = x^2$.

Desmos Tip: Graph $y = (x^{\frac{4}{3}} \cdot x^{\frac{9}{2}})/x^{\frac{1}{6}}$ and $y = x^2$ for $x > 0$ on Desmos and confirm the two curves lie exactly on top of each other.